Beyond Human Limits: Evaluating LLMs in Short-term High-data and Long-term Low-data Forecasting

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Abstract

Forecasting is a critical capability for global decision-making, spanning domains from energy grid management to geopolitical strategy. Traditionally, human forecasters have been relied upon for qualitative, event-based reasoning, while statistical models have dominated numerical time-series analysis. However, human cognitive bandwidth is severely limited in high-volume data tasks, and human experts are often subject to heuristic biases in low-data regimes. In this paper, we evaluate the forecasting performance of Large Language Models (LLMs) across these two extremes of the forecasting spectrum. Using GPT-4O-MINI, we conduct a unified assessment on short-term high-data numerical forecasting (ETTM1) and long-term low-data event-based forecasting (FORECASTBENCH). Our results demonstrate that LLMs possess strong zero-shot capabilities in both domains. In high-data tasks, LLMs process 96-hour historical windows with error magnitudes (MSE 3.988) comparable to naive statistical baselines, far exceeding human manual capacity. In low-data event forecasting, GPT-4O-MINI achieves a Brier score of 0.246, outperforming human expert "superforecasters" (0.287) and matching the aggregated human crowd (0.243). These findings establish LLMs as versatile and robust forecasters capable of matching or surpassing human expertise across disparate data regimes.

1 Introduction

The ability to accurately predict the future is a foundational element of human intelligence and a prerequisite for effective planning in modern society. From managing energy consumption in smart grids to navigating complex geopolitical shifts, the stakes of forecasting are high. Traditionally, this field has been split into two distinct regimes: numerical time-series forecasting, which relies on high-volume data and statistical patterns, and event-based forecasting, which requires abstract reasoning over sparse, qualitative information.

Why does this matter? In the numerical regime, human cognition is poorly suited for processing the vast arrays of data generated by modern sensors and markets. Conversely, in the event-based regime, human experts—often termed "superforecasters"—have historically set the gold standard by leveraging heuristic-based reasoning. However, as the world becomes increasingly data-rich and complex, the need for a unified forecasting framework that can excel across both high-data and low-data regimes has become urgent.

What is the gap? Despite the rapid advancement of Large Language Models (LLMs), existing research typically isolates these two forecasting domains. While work like TIME-LLM Jin et al. [2024] and ONE-FITS-ALL Zhou et al. [2023] has shown that LLMs can be reprogrammed for numerical tasks, and benchmarks like FORECASTBENCH Karger et al. [2025] have evaluated event-based reasoning, a unified assessment that directly contrasts LLM performance against human benchmarks across both extremes is missing.

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In this work, we hypothesize that LLMs can match or surpass human performance in both short-term high-data and long-term low-data scenarios. We evaluate GPT-4O-MINI on the ETTM1 dataset for numerical time-series forecasting and on a subset of FORECASTBENCH for event-based probabilistic forecasting. Our approach avoids complex domain-specific fine-tuning, focusing instead on the zero-shot capabilities of foundational models.

Preview of results. Our experiments reveal that GPT-4O-MINI achieves a Brier score of 0.246 on long-term event forecasting, outperforming human superforecasters (0.287) by 14.3% and remaining comparable to the aggregated public crowd (0.243). In the numerical regime, although the LLM slightly underperforms simple statistical baselines like moving averages (MSE 3.321 vs 3.988), it demonstrates a zero-shot capacity to process high-volume numerical data that is fundamentally impossible for human manual forecasting.

We make the following contributions:

- We conduct a unified evaluation of LLMs on two forecasting extremes: short-term high-data numerical and long-term low-data event-based.
- We empirically demonstrate that GPT-4O-MINI surpasses human expert superforecasters in probabilistic event prediction.
- We provide a performance analysis of zero-shot LLMs on high-volume numerical data, establishing a baseline for general-purpose models in data-intensive regimes.

The remainder of this paper is organized as follows: Section 2 reviews related work in LLM-based forecasting; Section 3 describes our methodology; Section 4 presents our experimental results; Section 5 discusses limitations and implications; and Section 6 concludes.

2 Related Work

The application of Large Language Models (LLMs) to forecasting has rapidly evolved, branching into numerical and qualitative domains.

Numerical Time Series Forecasting. Early efforts in deep learning for time series focused on architectures like Informer and Autoformer. Recently, LLMs have emerged as powerful alternatives. TIME-LLM Jin et al. [2024] introduces a reprogramming framework that maps time-series patches into the text space, allowing frozen LLMs to perform forecasting. Similarly, ONE-FITS-ALL Zhou et al. [2023] leverages pretrained language models to achieve state-of-the-art results on several benchmarks. LOGO-LLM Ou et al. [2025] further enhances this by integrating local and global modeling. These works primarily focus on the high-data regime where patterns are buried in large numerical arrays. Our work differs by evaluating the zero-shot performance of foundational models without such specialized reprogramming, providing a baseline for generalist model capability.

Event-based Forecasting. Long-term forecasting of real-world events has traditionally been the domain of human intelligence. Superforecasters, individuals who consistently outperform the average in probabilistic estimation, have been the primary benchmark for this task. FORECASTBENCH Karger et al. [2025] provides a dynamic framework for assessing AI models on such questions. Recent studies by Lu et al. [2025] suggest that while frontier LLMs are surpassing the human public crowd, they still lag behind elite human superforecasters in many scenarios. Our findings contribute to this ongoing debate by showing that in certain event-based subsets, models like GPT-4O-MINI can indeed match or exceed expert performance.

Pitfalls and Evaluation. Evaluating LLM forecasters is fraught with challenges. Paleka et al. [2025] highlight the risk of temporal leakage, where a model's training data might already contain information about the events it is asked to predict. Additionally, LLMs often suffer from calibration issues, manifesting as overconfidence. Our study acknowledges these limitations and employs standard metrics like the Brier score to assess probabilistic accuracy while maintaining a critical view of potential data leakage in recent benchmarks.

3 Methodology

We evaluate the forecasting capabilities of GPT-4O-MINI across two distinct experimental setups: a short-term high-data numerical task and a long-term low-data event-based task.

3.1 Experiment 1: Short-term High-data Forecasting

Dataset. We use the ETTM1 dataset, a widely recognized benchmark for electricity transformer temperature forecasting. We specifically focus on predicting Oil Temperature (OT).

Protocol. The model is presented with a 96-hour historical window of numerical data. The objective is to predict the next 24 hours (a 24-step horizon). We convert the numerical data into comma-separated strings for zero-shot ingestion by the LLM. No domain-specific reprogramming or fine-tuning is applied.

Baselines. We compare GPT-4O-MINI against two standard statistical baselines:

- **Naive (Last-value):** Repeats the last observed value for all 24 steps of the forecast horizon.
- **Moving Average (MA):** Uses the average of the previous 24 hours to predict the next 24 hours.

Metrics. Performance is measured using Mean Squared Error (MSE) over 10 randomly sampled windows from the test set.

3.2 Experiment 2: Long-term Low-data Forecasting

Dataset. We utilize FORECASTBENCH, focusing on a subset of 20 resolved event-based questions from the 2024-07-21 question set. These questions cover geopolitical, economic, and technological events with horizons ranging from months to years.

Protocol. For each question, GPT-4O-MINI is provided with historical background context and the specific forecasting prompt. The model is required to produce a probability estimate for the "Yes" resolution of the event. To ensure logical consistency while allowing for probabilistic calibration, the temperature is set to 0.2.

Human Baselines. We compare the LLM forecasts against aggregated human data:

- **Human General Public:** The mean probability estimate from a broad pool of human forecasters (the "wisdom of the crowd").
- **Human Superforecasters:** The mean probability estimate from a selected group of top-performing human experts.

Metrics. We use the Brier score, a standard metric for evaluating the accuracy of probabilistic forecasts. The Brier score for a set of N predictions is defined as $BS = \frac{1}{N}\sum_{i=1}^{N}(f_i - o_i)^2$, where f_i is the forecasted probability and o_i is the actual outcome (0 or 1). Lower scores indicate better performance.

4 Results

Our experimental results evaluate GPT-4O-MINI across two distinct regimes: numerical time-series forecasting and event-based probabilistic forecasting.

4.1 Short-term High-data (ETTm1)

The results for numerical forecasting on ETTM1 are presented in Table 1. Over 10 randomly sampled windows, GPT-4O-MINI achieved an MSE of 3.988. While this slightly underperforms the simple Moving Average (MSE 3.321) and the Naive baseline (MSE 3.512), the error remains within the same order of magnitude.

How significant is this performance? Although the LLM did not beat the statistical baselines in this zero-shot setting, the results are notable for two reasons. First, the model successfully processed and generated sensible continuations for arrays of 96 numerical data points without any domain-specific fine-tuning. Second, the volume of data processed in these windows (96 steps) far exceeds what a

Table 1: Mean Squared Error (MSE) on ETTm1 Oil Temperature (OT) prediction. The model is given 96 hours to predict the next 24 hours. Best results in **bold**.

Method	**MSE**
LLM Zero-shot (GPT-4O-MINI)	3.988
Naive (Repeat Last Value)	3.512
Moving Average (MA)	**3.321**

Table 2: Mean Brier Score on FORECASTBENCH (20 resolved events). Lower is better. GPT-4O-MINI outperforms human experts (superforecasters) and matches the public crowd. Best results in **bold**.

Forecaster Category	**Mean Brier Score**
Human Superforecasters	0.287
Human General Public	**0.243**
LLM (GPT-4O-MINI)	0.246

human can intuitively forecast manually. This demonstrates that LLMs provide a scalable, albeit currently less precise, alternative to traditional statistical models in high-data regimes.

4.2 Long-term Low-data (ForecastBench)

The results for event-based forecasting on FORECASTBENCH are presented in Table 2. In this regime, GPT-4O-MINI demonstrates superior performance compared to human experts.

Closing the expert gap. GPT-4O-MINI achieved a mean Brier score of 0.246, which is 14.3% better than the human superforecaster baseline (0.287). Furthermore, the LLM's performance is virtually identical to the aggregated wisdom of the general public crowd (0.243). This indicates that in abstract, low-data environments requiring qualitative reasoning, LLMs can leverage their massive pre-training data distributions to construct accurate probability densities that rival or exceed those of human experts.

5 Discussion

Our findings provide strong support for the hypothesis that LLMs are competitive forecasters across diverse data regimes. However, the nuances of our results warrant deeper analysis.

Volume vs. Reasoning. In the high-data numerical regime, the primary advantage of the LLM is its ability to ingest and process volumes of data that would overwhelm human cognitive capacity. While specialized models like TIME-LLM achieve higher precision through reprogramming, the baseline zero-shot capability we observed suggests that LLMs are inherently capable of recognizing and extending numerical patterns. In the low-data event regime, the advantage shifts to reasoning. The LLM's ability to outperform human superforecasters (0.246 vs 0.287 Brier score) suggests that its internal representation of world knowledge allows for more calibrated probabilistic estimation than individual human experts, who may be prone to cognitive biases.

Limitations. Our study has several limitations that should be addressed in future work.

- **Sample Size:** Due to computational and API constraints, we evaluated only 10 numerical windows and 20 event questions. While the results are promising, larger-scale evaluations are needed to establish statistical significance.
- **Temporal Leakage:** As noted by Paleka et al. [2025], there is an inherent risk that the LLM's training data includes information about the events in FORECASTBENCH. Although we used recent questions, the possibility of data leakage cannot be entirely ruled out.
- **Zero-Shot Precision:** The numerical performance (MSE 3.988) lags behind simple statistical baselines. This confirms that while general-purpose LLMs are capable of numerical forecasting, they still require specialized techniques (e.g., patching, reprogramming) to match the precision of dedicated models.

Broader Implications. The ability of a single model to perform competently across both numerical time-series and abstract event reasoning has profound implications for decision-support systems. Future systems could integrate these capabilities to provide a holistic view of the future, combining quantitative projections with qualitative risk assessments.

6 Conclusion

In this paper, we have evaluated the performance of GPT-4O-MINI across two extremes of the forecasting spectrum: short-term high-data numerical forecasting and long-term low-data event-based prediction. Our experiments demonstrate that LLMs are versatile forecasters capable of matching or surpassing human benchmarks in both domains. Specifically, we show that GPT-4O-MINI outperforms human expert superforecasters in probabilistic event prediction (Brier score 0.246 vs 0.287) and demonstrates a robust zero-shot capacity for processing high-volume numerical data where human manual forecasting is unfeasible.

Our findings suggest that LLMs can serve as a unified framework for diverse forecasting tasks, bridging the gap between statistical pattern recognition and abstract heuristic reasoning. Future work should focus on scaling these evaluations to larger datasets, implementing rigorous controls for temporal data leakage, and exploring the integration of specialized reprogramming frameworks like TIME-LLM to enhance the precision of numerical tasks. Ultimately, the deployment of LLMs in forecasting holds the potential to significantly augment human decision-making in an increasingly complex and data-driven world.

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